scl
Future Food Systems

WIREFRAMED BREAKDOWN FOR THE AGROTECH ECOSYSTEM

After a brief review of the client's operations and objectives, I have developed a system architecture that outlines the company's digital needs. The architecture is broken down into four core entities, each designed to be fully interoperable—ensuring seamless data exchange and operational synergy across all platforms.

Out of the four systems, three will include both mobile and web applications, while one will function solely as a web-based platform. This document provides an insight into how the systems are structured, how they interact, and how workflows will be executed across the ecosystem.

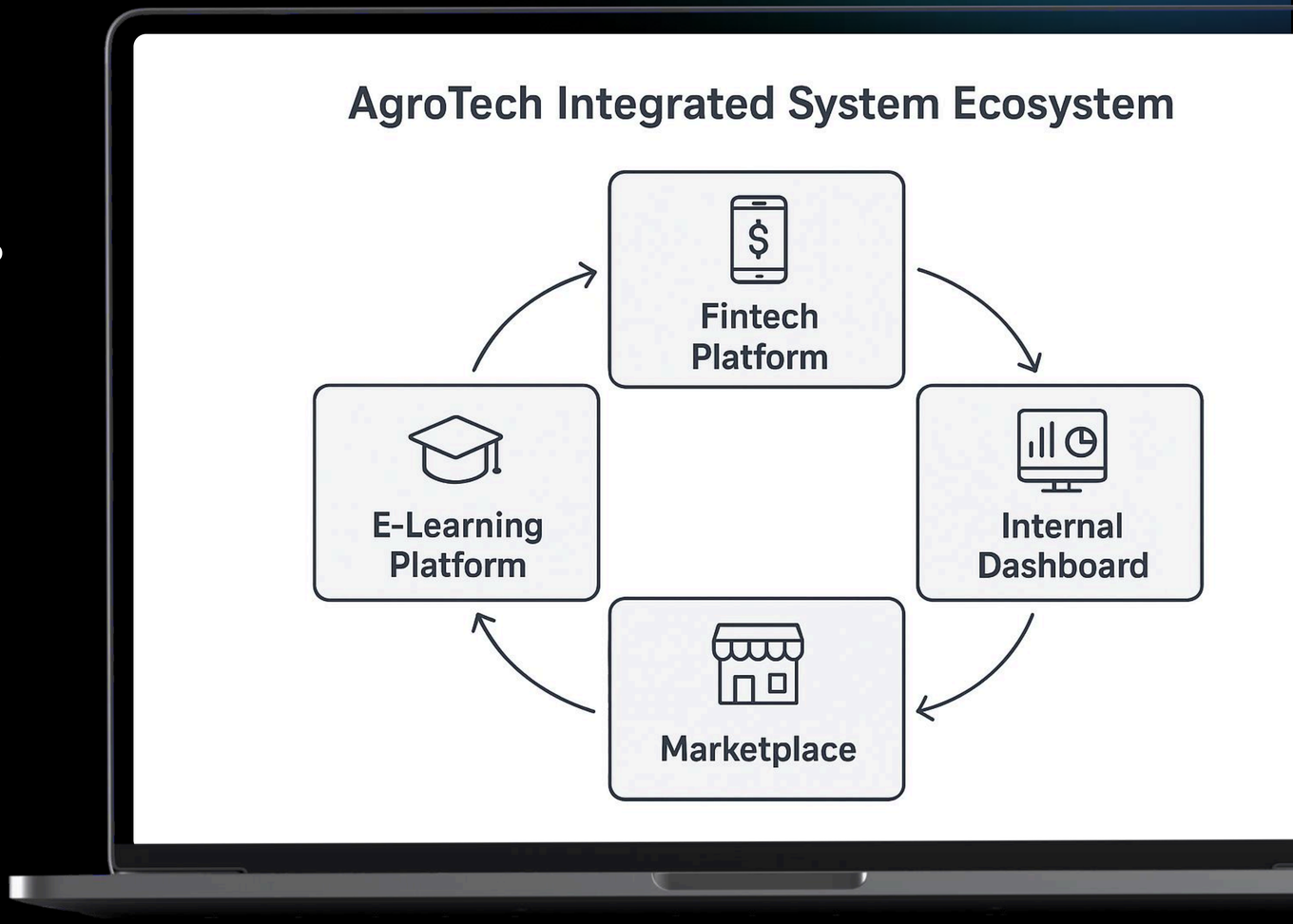
*Please note: This is not a final or approved design. It represents a conceptual system flow intended to illustrate the interaction, functionality, and architecture of the proposed solutions.

Ecosystem Breakdown

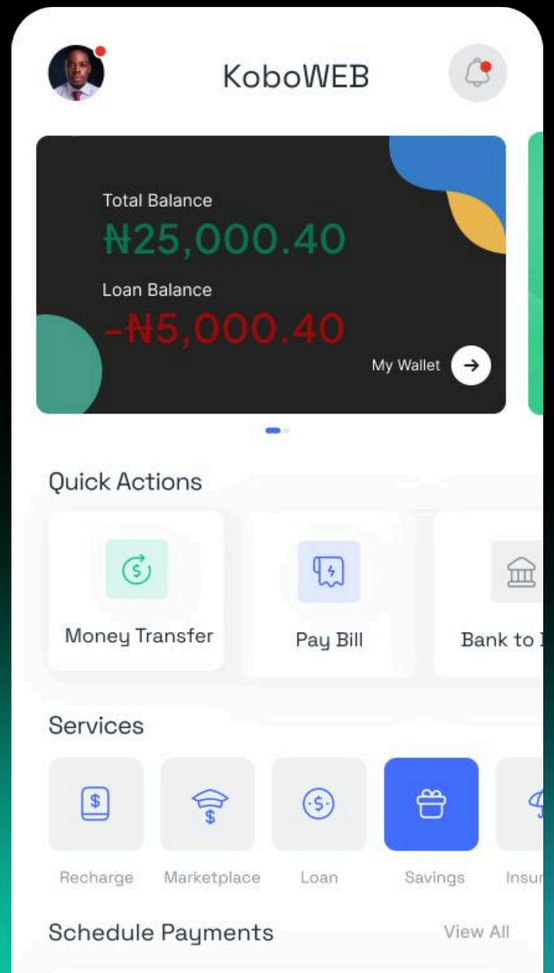
An interoperable suite of four intelligent digital systems designed to empower farmers, enhance productivity, and streamline operations across the agricultural value chain. Each system is built to address a specific functional area—finance, education, commerce, and administration—while collectively contributing to a unified ecosystem that promotes growth, transparency, and data-driven decision-making.

The platforms are fully synchronized through shared data APIs and centralized authentication (SSO), enabling seamless communication and interoperability between systems. This ensures that every farmer, vendor, and administrator interacts within a cohesive digital environment where information flows securely and efficiently.

By integrating financial technology, e-learning, digital marketplace solutions, and enterprise management tools, this ecosystem not only digitizes agricultural operations but also fosters knowledge sharing, financial inclusion, and sustainable development within farming communities.

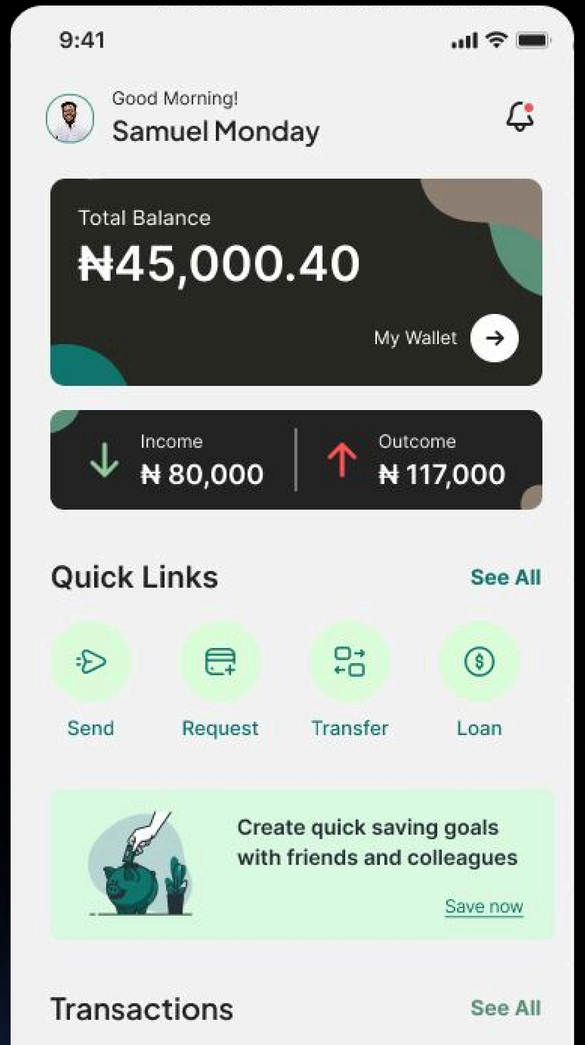


1. FINTECH SYSTEMS (KOBOWEB)



The current prototypes presented are not the final design but rather conceptual representations aimed at illustrating functionality and user flow. Our goal in the final design phase is to create a platform that is highly user-friendly and accessible, especially considering that the primary users are farmers who may have varying levels of digital literacy.

At the same time, we aim to ensure the app maintains a sleek, ultra-modern look and feel, reflecting the company's innovative vision and commitment to digital transformation in agriculture. The final design will balance simplicity and sophistication, ensuring a smooth user experience without compromising on aesthetics or usability.



💰 Fintech Platform – “Kobo” (Web & Mobile)

1. Purpose & Core Concept

Kobo serves as the **financial backbone** of the AgroTech ecosystem.
It enables farmers to **save, borrow, invest, and make transactions** in a secure, agriculturally tailored digital banking environment.
The platform integrates seamlessly with the **Marketplace**, **E-Learning**, and **Internal Dashboard**, creating a unified agricultural finance network.

2. User Roles

Role	Description
Farmer (Primary User)	Manages wallet, savings, loans, and payments.
Admin (Internal Staff)	Oversees transactions, approves loans, and monitors system health.
Agent (Field Officer)	Assists farmers with onboarding, KYC, and transactions.
Vendor (External Partner)	Receives payments for goods/services through Kobo.

System Flow (Simplified Overview)

```
Farmer → Registers & completes KYC → Creates Wallet
        → Saves / Borrows / Transacts → Uses Kobo Wallet for:
              ↳ Marketplace Purchases
              ↳ E-Learning Incentives
Admin → Monitors transactions via Dashboard
Data → Synced with Internal Dashboard (real-time analytics)
```

CORE FEATURES FOR KOBOWEB

💳 Digital Wallet & Payments

MVP

💰 Savings & Investment Plans

MVP

🌾 Agricultural Loans & Credit Access

MVP

📊 Financial Dashboard & Analytics

MVP

🔗 Ecosystem Integration

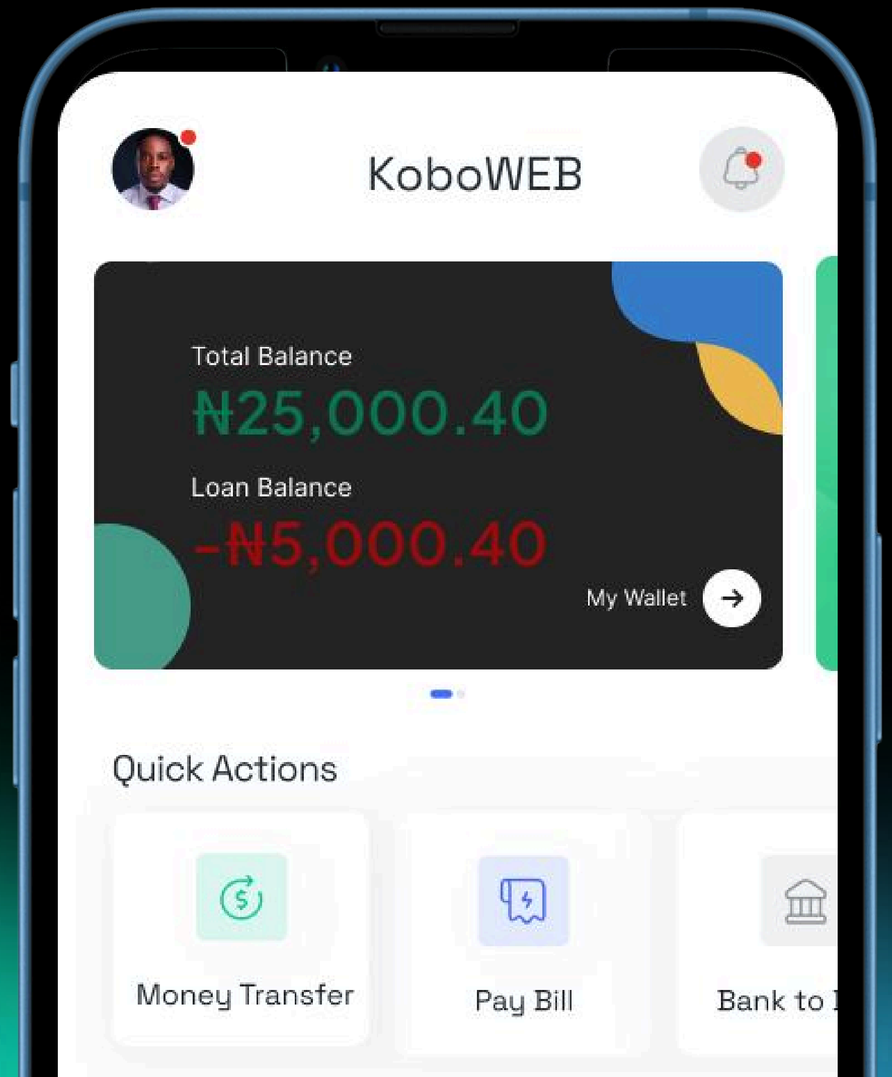
FFP

🛡️ Secure KYC & Compliance System

MVP

Mobile Application Koboweb

Key Features



A. Simple Onboarding & KYC
- Quick sign-up via phone number or email.
- Offline KYC collection by agents (auto-syncs when online).
- Multi-language support (English + local dialects).

B. Wallet & Transactions
- View balance and transaction history.
- Add funds via bank transfer, card, or agent deposit.
- Peer-to-peer transfers within Kobo network.
- Withdraw funds to bank or mobile money account.

C. Savings Plans
- Individual and group savings.
- Goal-based targets (e.g., “buy fertilizer in 3 months”).
- Visual progress bars for motivation.

D. Loan Access
- Loan application through guided steps.
- Automatic credit scoring (based on E-Learning activity + payment history).
- Notifications for approvals, disbursements, and repayments.

E. Marketplace Integration
- Pay for inputs or equipment directly from wallet.
- Track credit-based purchases and repayment schedules.

F. Rewards & Gamification
- Earn wallet bonuses or credits for completing E-Learning courses.
- Streaks for consistent saving or repayment habits.

G. Notifications & Alerts
- Push notifications for payments, reminders, or new loan offers.
- SMS fallback for low-data users.

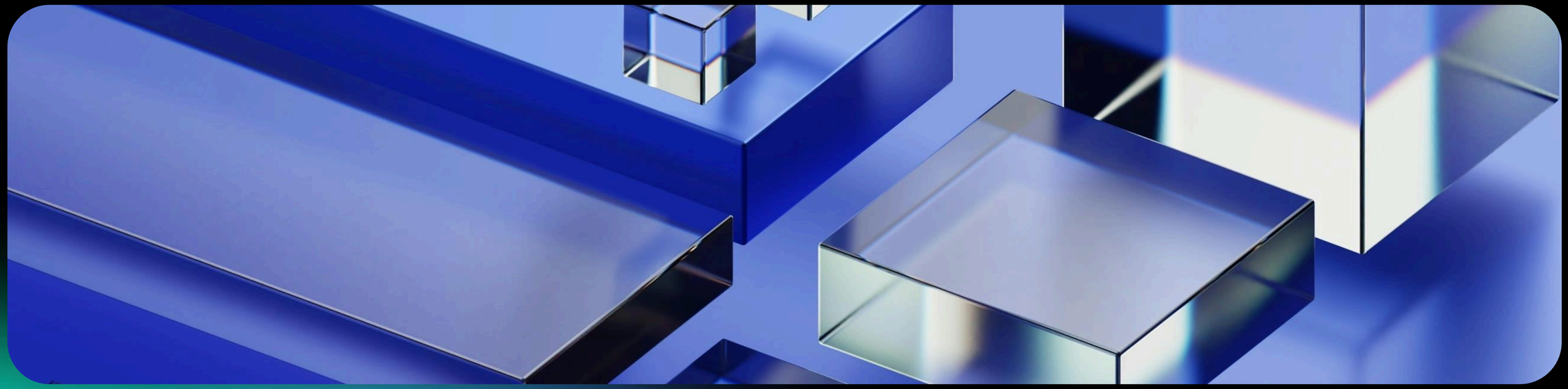
FINTECH SYSTEMS (KOBOWEB)

Security Architecture

Beside is a clear, practical security architecture for Koboweb covering both internal protections (platform, admin, ops) and protections for end users (farmers, agents, vendors). It lists controls, how they fit together, and prioritized security features

Internally, the platform operates within a segmented cloud infrastructure where each layer—from the public interface to the private databases—is isolated and strictly monitored. All traffic between servers passes through an API gateway and web application firewall that filters malicious requests, prevents injection attacks, and mitigates denial-of-service attempts. Administrative access is restricted through a secured VPN and multi-factor authentication, ensuring that only authorized personnel can reach sensitive environments.

User authentication across the entire ecosystem is handled through a centralized identity service using OAuth 2.0 and OpenID Connect protocols, enabling single sign-on between the different systems. Access rights are controlled through a strict role-based access model that separates permissions for farmers, vendors, agents, and administrators.Sensitive user data, including identification numbers and payment information, is tokenized and encrypted. The system never stores card verification data and relies on a PCI-DSS compliant payment gateway to handle financial transactions. Every token used for authentication is short-lived, reducing the risk of session hijacking, and refresh tokens are stored securely. For users, additional layers of security such as two-factor authentication, PIN verification for high-value transactions, and biometric login options are included.



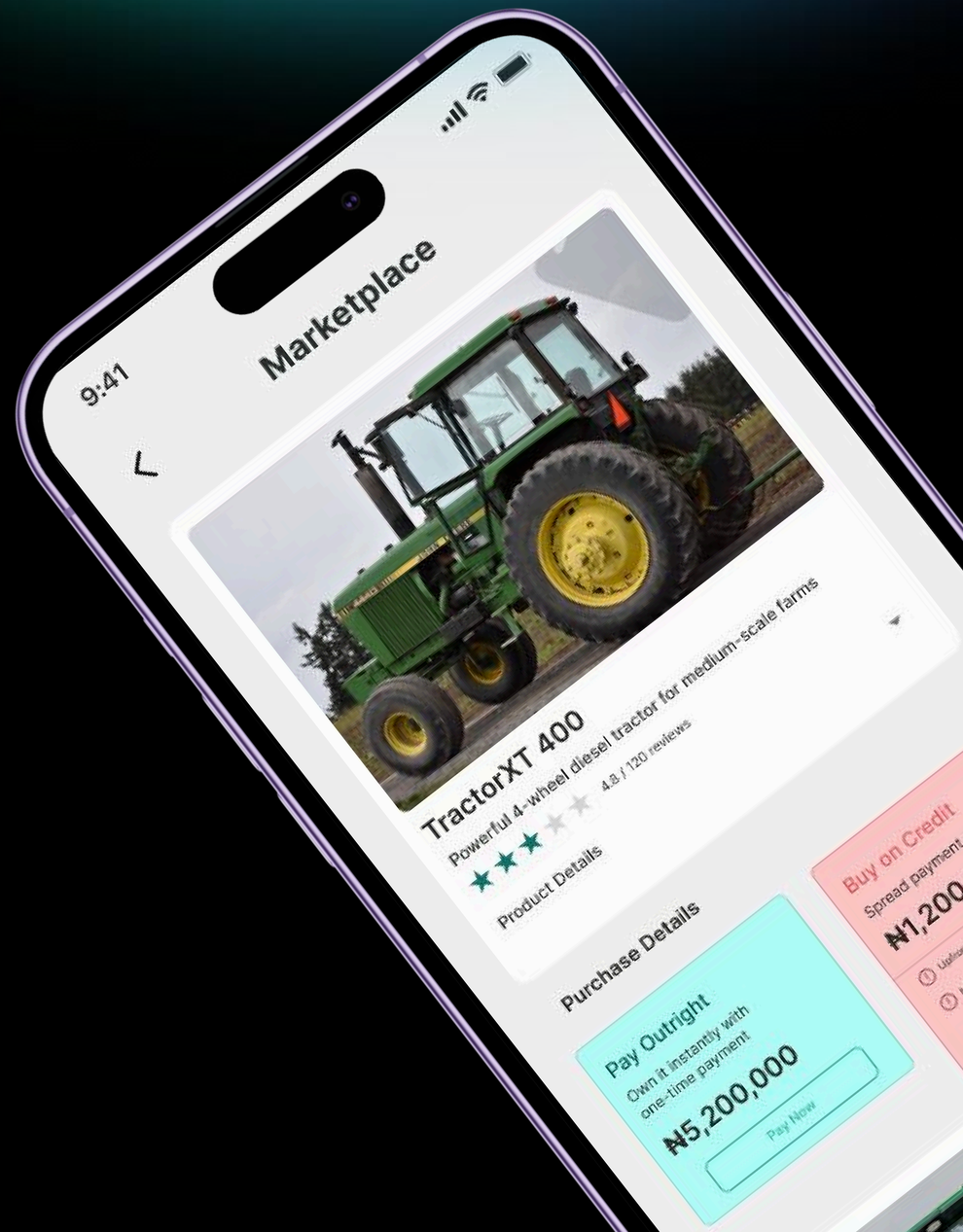
2. MARKETPLACE

The Kobo Marketplace is a digital agricultural ecosystem designed to connect farmers, suppliers, and consumers through a secure and transparent platform. It serves as a one-stop marketplace where verified agricultural products—from seeds and fertilizers to equipment and processed goods—can be bought, sold, and financed seamlessly. The platform bridges the gap between rural producers and urban buyers, ensuring fair pricing, product authenticity, and reliable delivery within a trusted digital environment.

At its core, the marketplace is structured to empower farmers and agribusinesses by giving them direct access to a wider customer base without the traditional intermediaries that often reduce their profit margins. Every seller on the platform undergoes a verification process to guarantee product authenticity and build trust across the value chain. This creates a more reliable ecosystem where buyers can confidently make purchases knowing that all products meet quality and safety standards.

The platform also introduces an innovative co-buying model that allows multiple users to pool funds together to purchase bulk items such as fertilizers or machinery at reduced prices. This feature is particularly beneficial for smallholder farmers who may not have the capital to make large purchases individually but can collectively gain access to the same resources as larger farms.

In addition, Kobo Marketplace integrates flexible buy-on-loan options through its fintech infrastructure. Users can apply for short-term financing directly within the platform to buy essential inputs or equipment. The loan process is streamlined and data-driven, using each user's transaction history and verified identity to determine eligibility. Repayments are automated and can be scheduled based on harvest cycles, making credit access realistic and sustainable for farmers.



CORE FEATURES FOR THE MARKETPLACE

🌾 Verified Product Listings

MVP

🤝 Co-Buying System

MVP

💰 Buy-on-Loan Option

MVP

💳 Koboweb Payment Integration

MVP

🚚 Product Delivery & Tracking

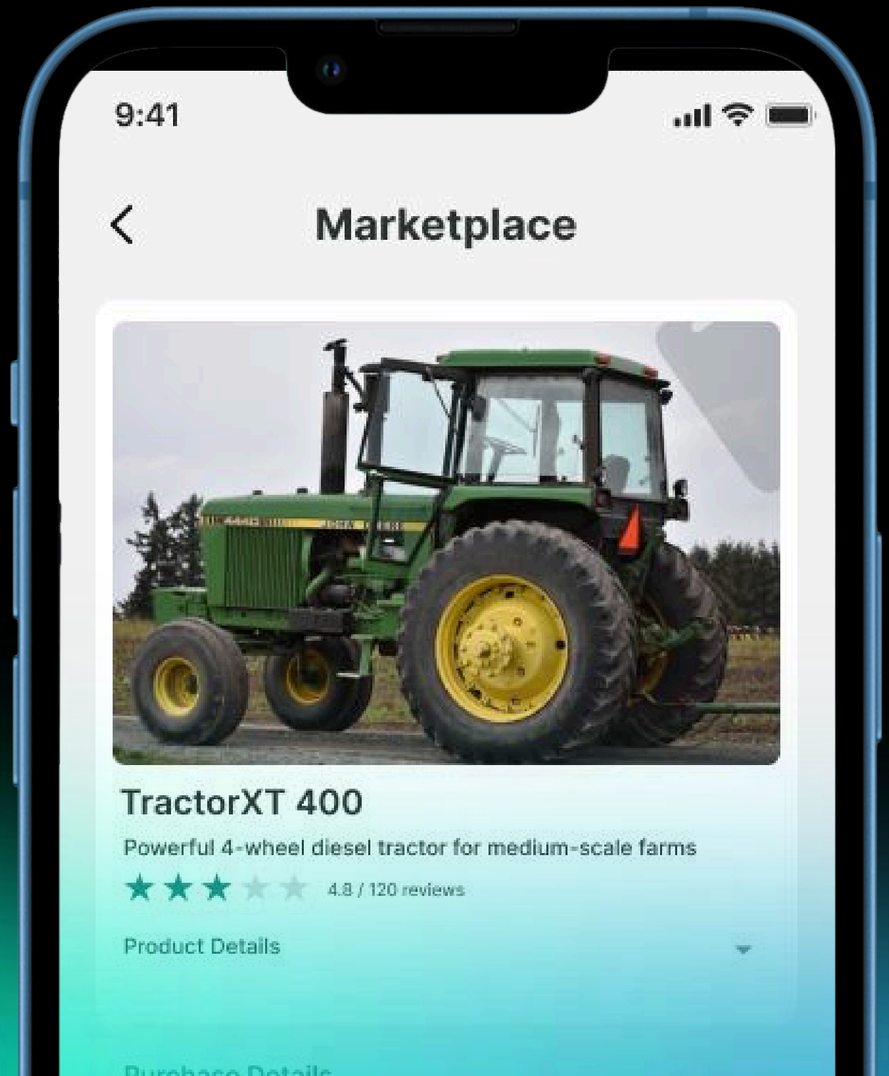
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🏬 Warehousing System Structure

FFP

Mobile Application Marketplace

Key Features



A. Verified Product Listings

All sellers and products are verified to ensure authenticity and quality.

Farm inputs, tools, and produce listings are categorized for easy discovery.

Seller ratings and buyer reviews help maintain marketplace trust.

B. Co-Buying System

Multiple users can pool funds to purchase bulk items collectively.

Co-buy groups are auto-matched by product type or region.

Discounted pricing tiers unlock based on group size.

C. Buy-on-Loan Option

Integrated with Kobo's fintech system for instant credit access.

Users can apply for short-term financing directly in the app.

Repayments are flexible—linked to harvest cycles or agreed terms.

D. Warehousing & Distribution Structure

Regional warehouses act as collection and distribution hubs.

Suppliers can store inventory for faster last-mile delivery.

System tracks inventory levels, expiry dates, and fulfillment status.

E. Payment & Settlement System

Supports Kobo Wallet, bank transfer, and card payments.

Harvest-based repayment options for loan purchases.

Automated settlements between sellers and buyers for transparency.

F. Logistics & Delivery Tracking

End-to-end delivery monitoring from warehouse to farm.

Integration with verified third-party logistics providers.

Real-time tracking and notifications for every order.

G. User Experience & Accessibility

Simple, intuitive interface designed for farmers.

Multi-language support (English + local dialects).

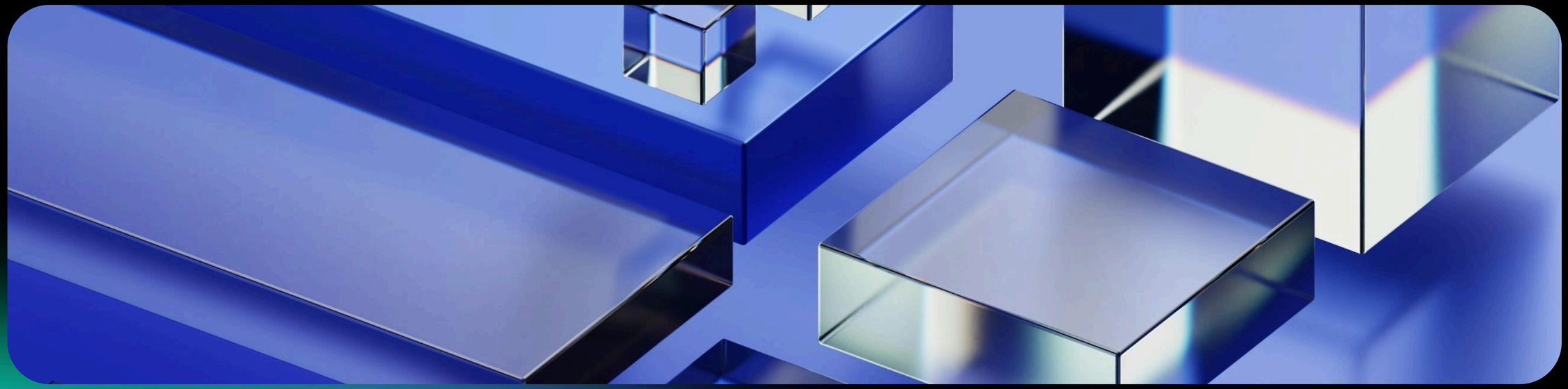
Responsive design across mobile and web platforms

MARKETPLACE

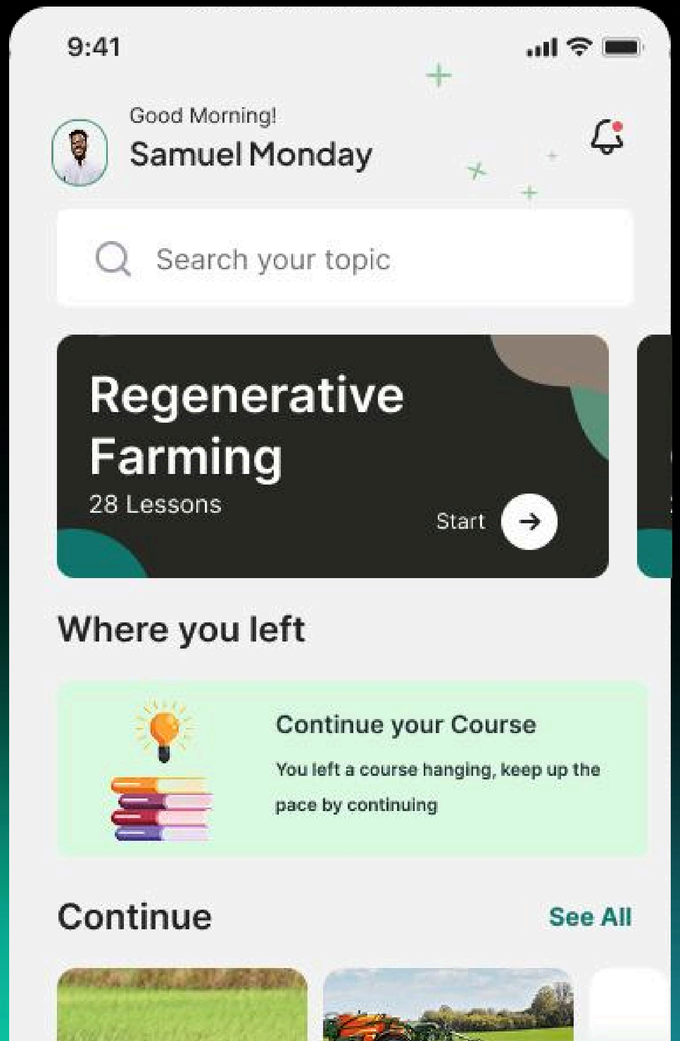
Security Architecture

Beside is a clear, practical security architecture for the marketplace covering both internal protections (platform, admin, ops) and protections for end users (farmers, agents, vendors). It lists controls, how they fit together, and prioritized security features

The security architecture for the Kobo Marketplace is built to protect product integrity, financial transactions, user data, and logistics operations across both digital and physical touchpoints. At its core the Marketplace enforces secure, authenticated access to every service and integration point, ensuring that vendors, buyers, warehouse operators, and logistics providers all interact through trusted channels. All communications between mobile apps, web clients, microservices, and third-party systems use TLS encryption and a centralized API gateway that applies authentication, rate limiting, and request validation before any business logic runs. Tokens are short-lived and issued by a centralized identity service that supports OAuth2/OpenID Connect, enabling single sign-on and consistent role management across Marketplace, Kobo Wallet, and the admin dashboard. Vendor and product verification is protected end-to-end: onboarding requires identity verification and documentation that is stored securely and linked to an auditable vendor profile. Product provenance and listing data are cryptographically signed in the system ledger so that product details, certifications, and authenticity proofs cannot be tampered with silently. Image and document uploads are scanned and validated to reduce fraud, while a tamper-evident audit trail records all listing changes, price edits, and inventory movements.

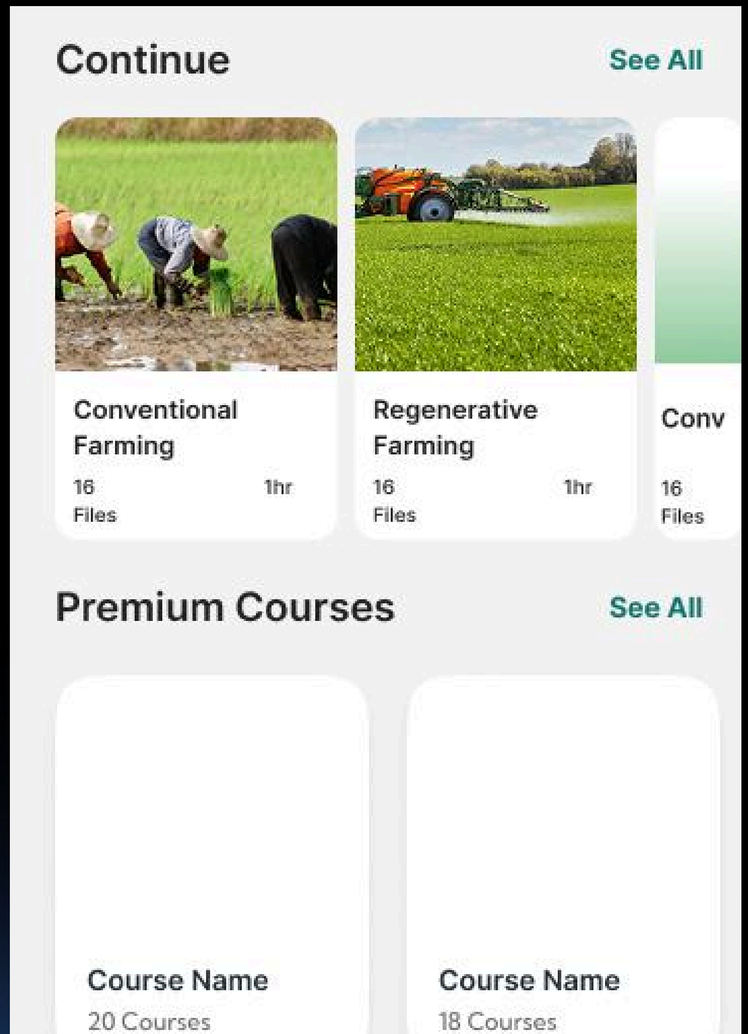


3. E-Learning Platform



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CORE FEATURES FOR THE MARKETPLACE

📚 Personalized Learning Dashboard

MVP

🎥 Multi-Format Educational Content

MVP

🏅 Certification & Skill Recognition

MVP

👩🏾‍🌾 Community Learning

MVP

🎮 Gamified Learning Experience

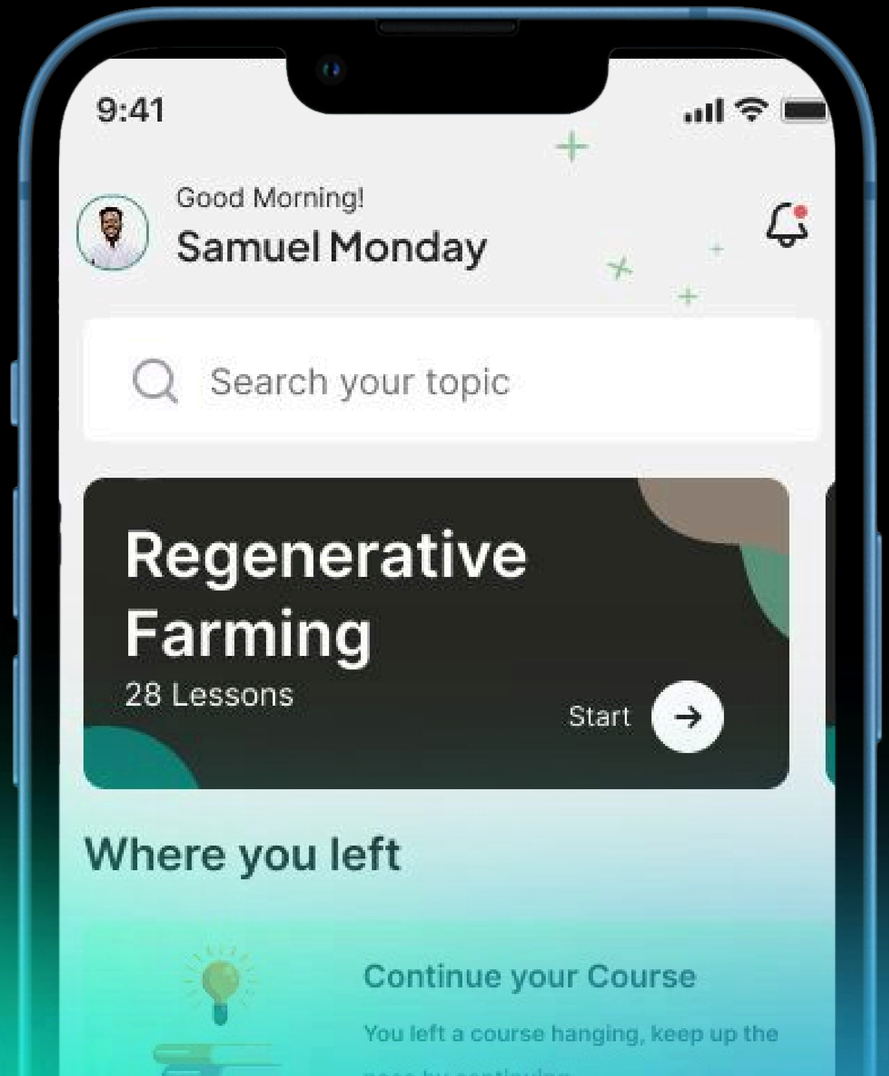
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🔗 Integration with Kobo Ecosystem

FFP

Mobile Application E-Learning

Key Features



A. Personalized Learning Dashboard

Each user has a tailored dashboard showing ongoing courses, progress, and recommendations.

Learning suggestions are generated based on farm type, region, and prior activity.

Offline mode allows users to continue lessons without internet access, syncing when online.

B. Multi-Format Learning Content

Courses are delivered through videos, audio lessons, infographics, and text modules.

Content is designed for different literacy levels, including voice-over narration and local language options.

Interactive simulations demonstrate modern farming techniques and equipment handling.

C. Certification & Skill Tracking

Learners receive digital certificates upon completing verified courses.

Badges and progress scores encourage continuous participation and skill advancement.

Certificates can be linked to user profiles in the Kobo Fintech app to improve credibility for loans or co-buy participation.

D. Community Learning & Peer Forums

Farmers can join community groups based on crop type or location.

Discussion forums and live Q&A sessions with experts encourage peer-to-peer learning.

Users can share farming experiences, success stories, and localized solutions.

E. Expert-Led Webinars & Mentorship

Regularly scheduled live sessions with agricultural experts, extension officers, and partner institutions.

Mentorship programs connect new farmers with experienced ones for ongoing support.

Recorded webinars remain accessible in the learning archive for continuous review.

F. Gamified Learning Experience

Courses integrate quizzes, rewards, and achievement milestones to motivate engagement.

Top-performing learners can earn tokens or discounts redeemable within the Kobo Marketplace.

Leaderboards promote healthy competition and community recognition.

Brand Guidelines



Typography across the ecosystem is consistent, utilizing Poppins as the primary font for modern readability and Inter for text-heavy interfaces that demand clarity. Imagery within the ecosystem is authentic and high-resolution, showcasing real farmers, farmlands, and produce. Illustrations are kept clean, flat, and minimal, ensuring communication remains clear and inclusive.

SCL stands for Sustainability, Connectivity, and Livelihood — the three core pillars that guide every design and strategic decision. The philosophy is built on the principle of empowering growth through technology, ensuring that every user experience feels natural, intuitive, and empowering.

The core color palette revolves around SCL Green, representing growth and agriculture; SCL Gold, symbolizing prosperity and opportunity; Charcoal Black, signifying depth and trust; and Pure White, ensuring clarity and simplicity.

Each platform within the ecosystem maintains its own subtle accent — the Fintech system adopts metallic tones that convey financial integrity, the Marketplace embraces earthy hues to evoke trade and productivity, the E-Learning system uses fresh pastel shades that stimulate creativity and openness, while the Dashboard maintains a neutral tone that encourages focus and precision.

How we make the Ecosystem Scalable

01.

Architectural Growth Strategy

Architecture

The SCL Ecosystem is built on a foundation of scalability, modularity, and interoperability — ensuring that as user adoption, data volume, and system complexity grow, performance and reliability remain uncompromised.

02.

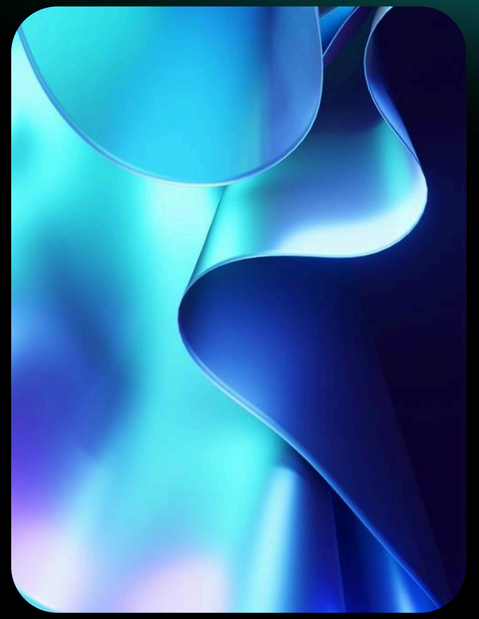
Conceptual / Strategic Perspective

Strategy

Each digital system — Fintech, Marketplace, E-Learning, and Internal Dashboard — was conceived as part of a modular framework that allows for continuous evolution. Instead of building one massive platform, the architecture separates core functions into distinct, interconnected components that can be independently expanded or optimized.

Key Core Requirements

Here we have strong technical section detailing the key core requirements of the SCL Ecosystem (applies to all four systems: Fintech, Marketplace, E-Learning, and Dashboard). It includes measurable data points like uptime, security level, response rate, and performance metrics



99.99%
Secure

Data security is a non-negotiable requirement. The system enforces end-to-end encryption (AES-256) for data at rest and TLS 1.3 for data in transit

>200ms
Load Rate

The ecosystem must support response times under 200 milliseconds for API requests and page load speeds under 2.5 seconds on standard 4G networks.

